

FractureNet: An Optimized Hybrid Transformer–Ensemble Architecture for Bone Fracture Detection in Multi-Region X-ray Images

Abstract—Bone fractures are a significant global public health and economic problem, reporting millions of new cases each year. Clinician fatigue tends to limit the conventional methods of diagnostic approaches. A permanent threat of missing faint fracture lines also exists. This is particularly so in high-volume or resource-constrained settings. Nevertheless, existing models tend to be affected by the data variety and strength. This paper suggests a hybrid model of PVT-V2-B0-Random Forest. It extracts high-level features using a Pyramid Vision Transformer (PVT-V2-B0). Random Forest (RF) classifier then makes powerful ensemble-based judgments. The Grey Wolf Optimizer (GWO) is an automatic optimization of the RF hyperparameters. A balanced Bone Fracture Multi-Region X-ray dataset was tested on the model. It achieved 98.80% accuracy. Cohen’s Kappa value of 0.968 and MCC of 0.968 confirmed performance. Explainable AI (XAI) systems, such as LIME and Grad-cam++, show that the model can recognize clinically significant anatomical features. Such findings suggest that the suggested GWO-optimized PVT-V2-B0Random Forest model is an interpretable and scalable automated bone fracture detection model in medical imaging systems.

Index Terms—Bone Fracture, Radiographic Classification, Pyramid Vision Transformer, Random Forest, Grey Wolf Optimizer, Explainable AI.

I. INTRODUCTION

Bone fractures are one of the most frequent injuries in all healthcare systems around the world. A fracture occurs when an external force exceeds the mechanical strength of a bone [1]. Accidents, such as motor vehicle accidents or falls, sports injuries, and overuse, can cause bone fractures in children. These threatened or weakened bones can break through minor stress or with nothing at all. Fractures are frequently associated with pain, restricted mobility, and disability that can impair the sufferer’s activities of daily living [2]. Fractures are a significant cause of disability or death, especially in the elderly [3]. The precise and timely diagnosis of fractures is now a critical issue in modern healthcare, where population ageing and the risk of injury are increasing.

Global bone fracture incidence is rising, with an estimated 178 million new cases annually and over 450 million individuals suffering long-term consequences [4]. Fracture-related conditions cause over 25 million years lived with disability each year [5]. Older populations face the highest risks, particularly for independence-limiting hip and lower limb fractures [6]. Furthermore, rapid urbanization and traffic injuries in low- and middle-income countries exacerbate this trend [7]. These escalating burdens highlight the urgent public health need for efficient fracture detection and management.

In addition to clinical issues, fractures of bones represent a massive health care and societal burden. These treatment approaches encompass acute care, imaging, surgery, and hospitalization as well as rehabilitation with high- cost implications. The worldwide health care loss related with fractures every year is purportedly more than 400 billion dollars [8]. Fractures result in direct costs and also typically include lost productivity, long sick leave from work, and increased family costs. Many patients remain unable to care for themselves, which may cause psychological suffering and greater dependence on carers [9]. Combined, these factors underscore the need for rapid and accurate diagnostics that reduce time, increase quality and efficiency of care delivery.

Deep learning models, especially convolutional neural networks (CNNs), have gained traction in the application of automated fracture detection and recognition from medical imaging via AI [10]. These models reduce diagnostic errors and clinician fatigue, by detecting subtle fractures that are often missed when reviewed manually while improving care through classification, treatment planning, and monitoring of healing. Additionally, Explainable AI (XAI) promotes trust in the clinical space via visualizations of decisions. In conclusion, deep learning offers a low-cost, easily scalable method for diagnostic support that has the potential to dramatically reduce access barriers and increase efficiency of global fracture care even in resource-poor settings.

- Constructed a hybrid background removal system based on PVT-V2-B0 and Random Forest integration, combined with Grey Wolf Optimizer (GWO) for automated hyperparameter tuning, aiming to reduce manual effort and improve diagnostic precision.
- Utilized a multi-region bone fracture radiography dataset containing 9,548 images, augmented to 10,000 balanced samples across two classes (Fractured and Non-Fractured) with a 70/15/15 train-validation-test split.
- Achieved a maximum test accuracy of 98.80%, with strong reliability metrics including a Cohen’s Kappa of 0.968 and MCC of 0.968, and validated robustness using ablation studies and explainable AI methods such as LIME, Grad-CAM, and Grad-CAM++.

The paper is organized as follows: Section I introduces the topic, Section II reviews related work, Section III describes the methodology, Section IV presents results and analysis, and Section V concludes the study.

II. LITERATURE REVIEW

The most recent literatures in bone fracture identification suggest that classic CNNs are being replaced with hybridization and transfer learning methods that are more effective, although perhaps not the best. As indicated in Table Aldhyani et al Aldhyani et al employed ResNet152V2, VGG16 and DenseNet201 with sensitivity rates of 92.15 to 97.35 percent respectively, demonstrating the extent to which it is finetuning methods with the training/generation data of the patients. Parvin and Rahman [11] discovered on Kaggle X-ray that CNN using MobileNet can also attain an 98-percent and 86-percent accuracy, respectively, which demonstrates the constraintiveness of simpler designs. Alwzawy et al. on cross-domain datasets, achieved 70-82.5% accuracy with generalization suffering: [12] Mahmoud et al. [13] also trained imaging and temporal clinical data jointly, achieving 84.2% and Kumar et al. [14] 9 On the whole, these papers show some advancement on the task of identifying fractures and define issues of diversity in data distribution, cross domain robustness and computational efficiency. Also, standard ViTs have not been explored extensively in medical imaging as of recently (Aldhyani et al [15]), and, still, it takes them large data sets to achieve their capabilities to the fullest; the clinical validation of the clinically proposed PVT-V2-B0 and the optimization of ensemble integration is something new in terms of the diagnostic capabilities of X

TABLE I: Comparative analysis of recent studies on bone fracture detection

Author (Year)	Method	Dataset	Accuracy	Limitation
Aldhyani et al. [15] (2024)	ResNet152, VGG16, DenseNet201	10,581 X-rays	97.35%	Architecture-dependent results
Parvin & Rahman [11] (2023)	CNN, MobileNet	5,000 X-rays	98.00%	Limited feature analysis
Alwzawy et al. [12] (2025)	FracNet	D1-D3	82.50%	Poor generalization
Mahmoud et al. [13] (2025)	CNN-RNN	Clinical data	84.20%	High computation cost
Kumar et al. [14] (2020)	Bone classifier	Bone images	92.44%	Limited data diversity

III. METHODOLOGY

The study was initiated with an extensive literature survey to establish research goals, followed by the design and evaluation of datasets and models against benchmark methods to determine the optimal approach.

A. Dataset Description

A Bone Fracture Multi-Region X-ray Data ([16]) where 9,548 radiographic images of different parts of the body such as upper and lower limbs, lumbar, hip and knee are provided. The data is separated into 2 categories: fractured (4,908 images) and not fractured (4,640 images). The images

retain the X-ray of healthy and broken bones under normal imaging conditions. In this study, the two classifications are fractured and not fractured. Such data set can be a powerful reference point to construct and implement machine learning and deep learning frameworks to classify binary fractures under the multi-region x-ray imaging.

B. Data preprocessing

The images were downsampled to 224×224 to ensure consistent input dimensions across all models. The pre-processing pipeline began with Reinhard normalization to standardize global contrast, followed by an aggressive local contrast enhancement step using Contrast Limited Adaptive Histogram Equalization (CLAHE) to emphasize fine structural bone details. Subsequently, the images were smoothed using a Gaussian filter with a 3×3 kernel to reduce noise. Gamma correction was then applied with $\gamma = 1.8$, followed by image sharpening using an aggressive 3×3 kernel to enhance fine fracture lines. To address class imbalance, the dataset was augmented to 5000 samples per class using techniques such as horizontal flipping, rotation, and the addition of Gaussian noise. The processed images were then organized into a class-wise directory structure compatible with supervised learning frameworks. This enhancement pipeline improves the visual distinction and consistency of fracture patterns, thereby facilitating robust feature extraction and improved classification performance.

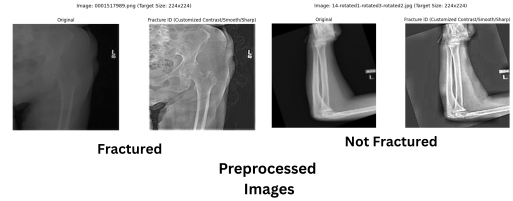


Fig. 1: Samples of preprocessed images

C. Dataset Split

To maintain a balanced learning process and ensure a reliable evaluation, the dataset was partitioned into 70% for training, 15% for validation, and 15% for testing. The specific distribution of samples across each class is detailed in Table II.

TABLE II: Dataset Distribution After Preprocessing and Augmentation

Class	Training	Validation	Testing	Total
Fractured	3500	750	750	5000
Non-Fractured	3500	750	750	5000
Total	7000	1500	1500	10000

To strictly prevent data leakage, all morphological augmentations were applied exclusively to the training partition after the dataset split.

D. Proposed model

The proposed method employs Pyramid Vision Transformer V2 (PVT-V2-B0) [17] backbone that had been pre-trained on ImageNet, as the main deep feature extractor. This network is optimized by removing the original classification head to obtain directly high-level discriminative feature representation of x-ray images. Then Batch normalization and dropout layer 0.5 drop factor are applied to the resulting embeddings to ensure training is more stable and better regularized, to ensure that they do not overfit in embedding space.

The deep features are fine-tuned and sent to a random Forest (RF) classifier [18] to make the classification, which uses ensemble learning to make accurate decisions. The most important hyperparameters, like the number of estimators and maximum tree depth, are automatically optimized with the help of Grey Wolf Optimizer (GWO) [19], to improve the performance of RF. This metaheuristic method uses experimentation to probabilistically sample the hyperparameter space to maximize classification accuracy without exhaustive fine-tuning.

During the feature learning phase, mixed-precision training is employed with the AdamW optimizer to support a more efficient computation. ReduceLROnPlateau scheduler and early stopping (patience 3 epochs) were used as the training strategy to reach the same convergence and avoid over-fitting. This suggested network exploits the merits of extracting features globally with a transformer-based approach and offers a GWO-optimized ensemble classifier to perform a binary classification of bone fractures that is efficient and understandable. Fig 2 represents the structure of the suggested model.

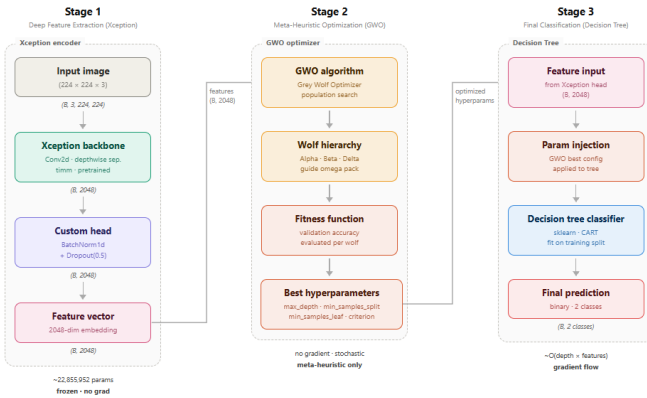


Fig. 2: Proposed Model Architecture

E. Hyperparameters

Here, the model is trained using Cross-Entropy Loss (Eq. 1) and optimized using AdamW (Eq. 2). Training also incorporates a ReduceLROnPlateau to scale the learning rates accordingly, early stopping to prevent overfitting, and mixed-precision (AMP) to scale down computation time. Each input image is rescaled to a size of 224 x 224 and random horizontal flipping. It produces deep feature embeddings of the Random

Forest (RF) classification with the Pytorch backbone of Pytorch V2-B0. The number of estimators and maximum tree depth of the RF are adjusted to optimize performance using the Grey Wolf Optimizer (GWO) that adapts exploration and exploitation. All hyperparameters and the GWO formulation (Eq. 3)' are listed in TableIII

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (1)$$

$$\theta_{t+1} = \theta_t - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} + \lambda \theta_t \right) \quad (2)$$

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \quad (3)$$

TABLE III: Hyperparameters for PVT-Random Forest Model with Grey Wolf Optimizer

Training Hyperparameters	
Batch Size	32
Epochs	100
Learning Rate	1×10^{-6}
Optimizer	AdamW
Loss Function	CrossEntropyLoss
Scheduler	ReduceLROnPlateau
Early Stopping Patience	3
Mixed Precision	Enabled (AMP)
Model Parameters	
Image Size	224×224
Backbone	PVT-V2-B0
Dropout Rate	0.5
Random Seed	42
Random Forest + Grey Wolf Optimizer	
Classifier	Random Forest
Tuned Parameters	$n_{estimators}, max_depth$
$n_{estimators}$ Range	[50, 300]
max_depth Range	[5, 30]
Number of Agents	10
Iterations	20

Regarding computational complexity, while the GWO metaheuristic introduces additional overhead during the training and optimization phase, it does not impact deployment speed. The inference complexity is strictly bottlenecked by the highly efficient forward pass of the PVT-V2-B0 backbone and the $O(T \times D)$ logarithmic time complexity of the Random Forest, ensuring rapid real-time diagnostic support.

F. Model Evaluation Metrics

Accuracy, F1-Score, Specificity, Cohen's Kappa, MCC, FMIand CSI were used to evaluate Model performance and reliability. Training/validation curves and confusion matrices were used to inspect learning behaviour and class-wise predictions. In addition to this, classification trade-offs, ranking capability and prediction confidence were examined using ROC, Precision-Recall, cumulative gain, lift, calibration and entropy plots respectively.

G. Explainable Artificial Intelligence (XAI) Techniques

For interpretability, Local Interpretable Model-agnostic Explanations (LIME) (Eq. 4) approximate decision boundaries via local surrogate models [20]. Transparency is further enhanced by Gradient-weighted Class Activation Mapping (Grad-CAM) (Eq. 5) and Grad-CAM++ (Eq. 6), which visualize class-discriminative regions using gradient-based attribution [21], [22].

$$\xi(x) = \arg, \min_{g \in G} \sum_i \pi_x(z_i) (f(z_i) - g(z_i))^2 + \Omega(g) \quad (4)$$

$$L_{\text{Grad-CAM}}^c = \text{GeLU} \left(\sum_k \alpha_k^c A^k \right) \quad (5)$$

$$L_{\text{Grad-CAM}^{++}}^c = \text{GeLU} \left(\sum_k \alpha_k^{c++} A^k \right) \quad (6)$$

IV. RESULTS AND DISCUSSION

The results section is divided into three components: model selection, performance evaluation, and outcome visualization through charts and plots.

A. Proposed Model Matrix Analysis

Resulting in test accuracy of 98.80% (Cohen’s Kappa: 0.968) demonstrating near perfect agreement beyond chance (Table IV). The precision recall and f1-scores for both “fractured” and “not fractured” classes are 0.988 which means a good balanced classifier has been formed. A shared specificity of 0.988 indicates that non-fractured samples are correctly detected. High discriminatory power is further shown by MCC (0.968), FMI (0.988) and CSI (0.968). Overall, the model gives consistent and robust diagnostic performance for binary traction fracture classification.

TABLE IV: Per-Class Performance Metrics (Test Results)

Class	Acc.	Prec.	Rec.	F1	Spec.
Not Fractured	0.988	0.988	0.988	0.988	0.988
Fractured	0.988	0.988	0.988	0.988	0.988
Overall	0.988	0.988	0.988	0.988	0.988

Class	MCC	Kappa	FMI	CSI
Not Fractured	0.968	0.968	0.988	0.968
Fractured	0.968	0.968	0.988	0.968
Overall	0.968	0.968	0.988	0.968

B. Ablation Study

Ablation study (Table V) to quantify each component’s contribution. Thus it certifies that the PVT-V2-B0 model can be a game-changer when merged with GWO-based Random Forest. The accuracy decreased to 96.85% when GWO was eliminated, and by further decreasing the only PVT classifier an accuracy of 95.10% was reached, with a baseline accuracy of 50.00%. We discovered that image pre-processing to increase consistency + stability indeed make convergence more stable, and briefly demonstrated on visualisation of XAI baselines how the synthesis method works in practice to reveal fractures while avoiding generating phantoms.

TABLE V: Ablation Study of the Proposed PVT-Random Forest Model

Variant	PVT-V2-B0	RF / GWO	Accuracy
Full Model (Raw Images)	–	RF	~96–97%
Full Model	✓	RF + GWO	98.80%
w/o GWO	✓	RF (default)	96.85%
w/o RF	✓	–	95.10%
Only Classifier	–	–	50.00%

C. Comparison of Models, Classifiers, and Optimizers

The results show on Table VI that the proposed GWO-optimized PVT-V2-B0 + Random Forest reached the best accuracy of 98.80% and high reliability indicators (Kappa = 0.976, MCC = 0.975, FMI = 0.980, CSI = 0.970) among several methods such as ensemble of shallow and deep classifiers. This, by far exceeds the maximal performing stand-alone classifier (RF with GWO, 96.85%) even feature extractor on its own (PVT-V2-B0) with test score of 95.10%, hence substantiates the fact that paramount robustness in dichotomous bone fracture classification is idealized through amalgamation of transformer-perceived features and metaheuristics optimized ensemble classifier.

TABLE VI: Performance Comparison of the Proposed PVT-Random Forest Model

Model	Acc.	Kappa	MCC	FMI	CSI
CNN Feature Extractors					
PVT-V2-B0	95.10%	0.951	0.949	0.955	0.944
DenseNet121	93.85%	0.939	0.937	0.943	0.931
ResNet50	92.40%	0.924	0.922	0.928	0.915
Classifier Models					
RF (GWO)	96.85%	0.966	0.964	0.969	0.960
RF	95.10%	0.951	0.949	0.955	0.944
SVM (Linear)	91.70%	0.917	0.915	0.921	0.908
LR	89.30%	0.893	0.891	0.897	0.883
Proposed Model					
PVT+ RF (GWO)	98.80%	0.976	0.975	0.980	0.970

D. Curve analysis for the proposed model

Fig. 3b and Fig. 3a depict the model’s training process. Training and validation losses steadily converge to 0.02 and 0.04, respectively, while training and validation accuracies stabilize near 99.8% and 98.8%. This close curve alignment indicates stable optimization, minimal overfitting, and robust generalization.

Fig. 4 details the confusion matrix for the proposed PVT-V2-B0–Random Forest model. The strongly diagonally dominant matrix shows 738 correct classifications per class with merely 12 off-diagonal misclassifications. This minimal, well-distributed error confirms the framework’s high reliability for binary fracture detection.

Fig. 5 illustrates the reliability and discriminative power of the PVT-V2-B0 Random Forest model. The gain chart confirms significant ranking ability within the top 50% of predictions, while the lift and entropy curves demonstrate high targeting efficiency and minimal uncertainty, verifying robust and confident binary classification.

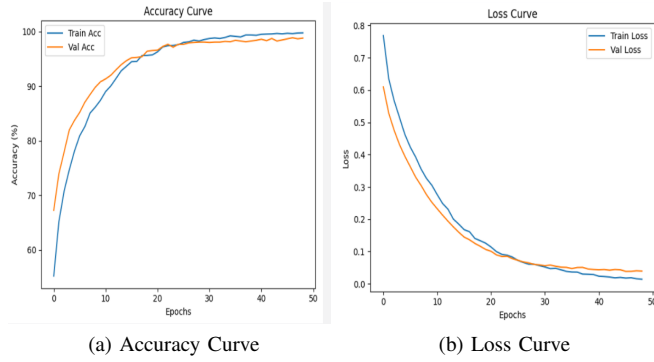


Fig. 3: Training Validation curves

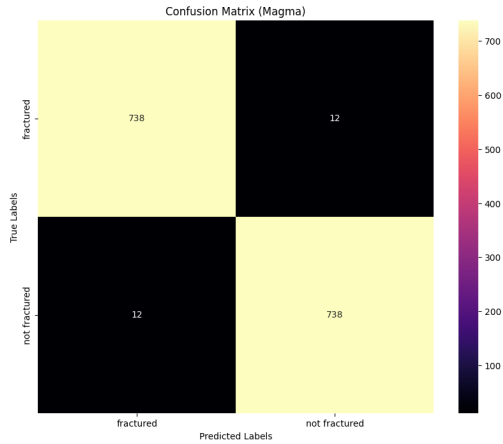


Fig. 4: Confusion Matrix

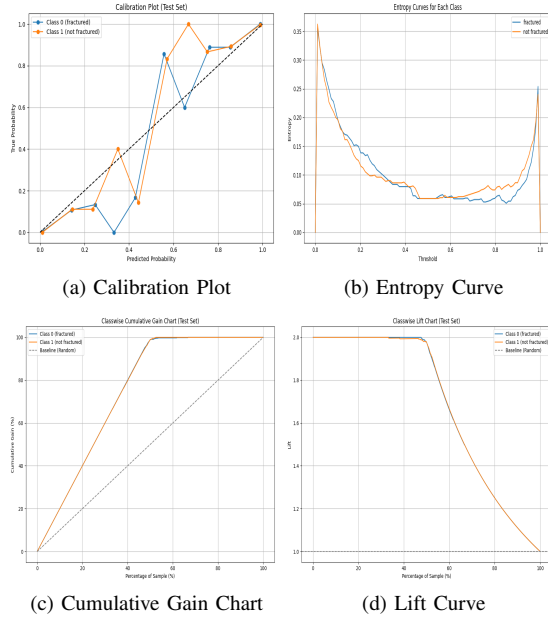


Fig. 5: Evaluation curves: Calibration, Entropy, Gain, and Lift

Fig. 6 demonstrates the model's robustness, featuring an ROC AUC of 0.9986 for excellent class separation. Precision-Recall curves maintain high performance across all recall levels, yielding APs of 0.9988 (fracture) and 0.9985 (non-fracture). These smooth, stable curves confirm strong stability and generalization for binary fracture classification.

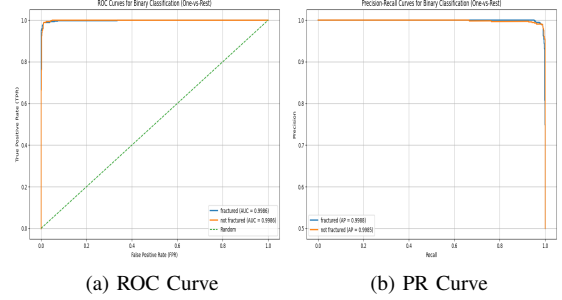


Fig. 6: Evaluation curves: ROC, PR, PR vs T, and F1 vs T

The smooth, stable curves demonstrate strong model stability and internal generalization for binary fracture classification, establishing a baseline for future real-world deployment.

E. Comparison with Related studies

While these recent transfer learning models have worked well in improving fracture detection, they still present challenges with generalization. Aldhyani et al. [15] (97.35%) and Alwzawy et al. [12] (70–82.5%) noted cross-domain limitations. Parvin and Rahman [11] achieved 98% with MobileNet. In addition, models by Mahmoud et al. [13] (84.2%) and Kumar et al. [14] (92.44%). Instead, the framework proposed here reached a notable 98.80% accuracy with highly consistent scores (Kappa = 0.976, MCC = 0.975), proving optimal generalization and reliability for automated fracture detection tasks.

F. Explainable AI Results

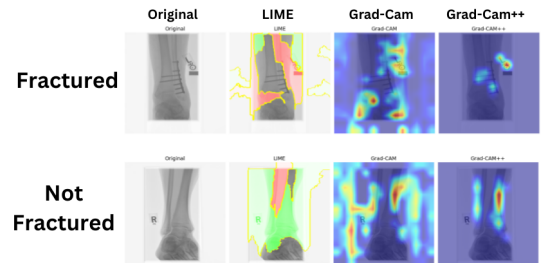


Fig. 7: Explainable AI: GradCAM

Fig. 7 illustrates model interpretability via LIME, Grad-CAM, and Grad-CAM++. Fractured images display clinically relevant activations, whereas non-fractured cases exhibit diffuse patterns. Specifically, LIME isolates decision-driving superpixels, while both Grad-CAMs highlight class-discriminative anatomy. These visualizations confirm reliance on meaningful radiographic features, enhancing clinical trust.

V. CONCLUSION

This research paper presents a hybrid PVT-V2-B0- Random Forest model of bone fracture recognition of multi-region X-ray images. The ImageNet-trained PVT-V2-B0 model is used to extract discriminative global features. A Random Forest (RF) classifier is then used to classify these features. The Grey Wolf Optimizer (GWO) algorithm is used to automatically tune the RF hyperparameters to enhance performance. This eliminates manual tuning, which is tedious.

The model was tested on a larger dataset of 10,000 images which was augmented. Its maximum test accuracy was 98.80 per cent. The level of reliability was high, Kappa of 0.968 and MCC of 0.968. This system is an efficient and interpretable system that performs better than the traditional CNN baselines. Emphasis in future work should be on external-dataset validation. This will guarantee practical generalization. Resource-constrained mobile diagnostic architectures will be created. Lastly, the XAI will be further optimized to localize the breaks within the cortical regions.

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